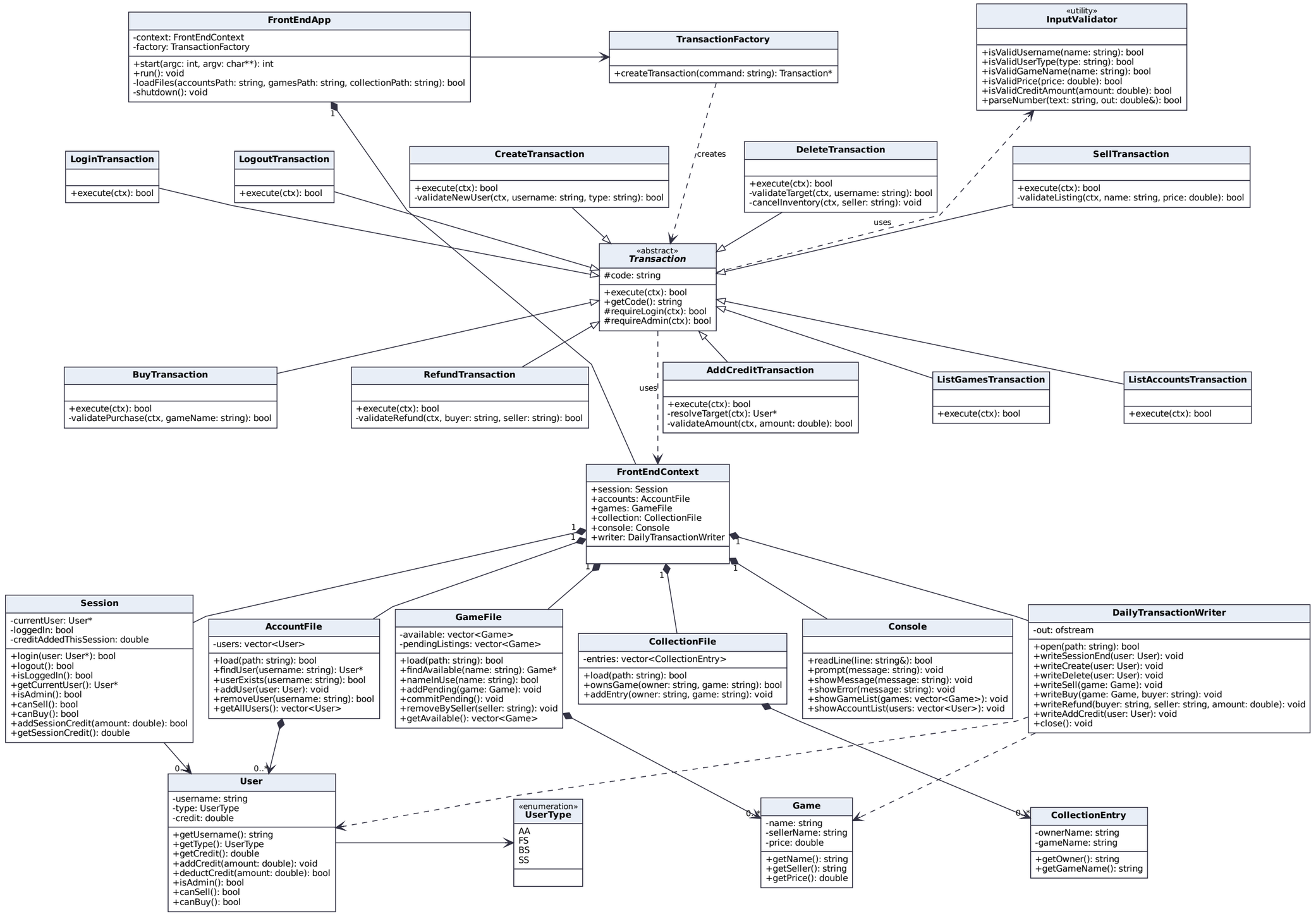

FrontEndApp
-context: FrontEndContext
-factory: TransactionFactory
+start(argc: int, argv: char**): int
+run(): void
-loadFiles(accountsPath: string, gamesPath: string, collectionPath: string): bool
-shutdown(): void
TransactionFactory
+createTransaction(command: string): Transaction*
«utility»
InputValidator
+isValidUsername(name: string): bool
+isValidUserType(type: string): bool
+isValidGameName(name: string): bool
+isValidPrice(price: double): bool
+isValidCreditAmount(amount: double): bool
+parseNumber(text: string, out: double&): bool
LoginTransaction
+execute(ctx): bool
LogoutTransaction
+execute(ctx): bool
CreateTransaction
+execute(ctx): bool
-validateNewUser(ctx, username: string, type: string): bool
DeleteTransaction
+execute(ctx): bool
-validateTarget(ctx, username: string): bool
-cancelInventory(ctx, seller: string): void
SellTransaction
+execute(ctx): bool
-validateListing(ctx, name: string, price: double): bool
creates
uses
«abstract»
Transaction
#code: string
+execute(ctx): bool
+getCode(): string
#requireLogin(ctx): bool
#requireAdmin(ctx): bool
BuyTransaction
+execute(ctx): bool
-validatePurchase(ctx, gameName: string): bool
RefundTransaction
+execute(ctx): bool
-validateRefund(ctx, buyer: string, seller: string): bool
AddCreditTransaction
+execute(ctx): bool
-resolveTarget(ctx): User*
-validateAmount(ctx, amount: double): bool
ListGamesTransaction
+execute(ctx): bool
ListAccountsTransaction
+execute(ctx): bool
uses
1
FrontEndContext
+session: Session
+accounts: AccountFile
+games: GameFile
+collection: CollectionFile
+console: Console
+writer: DailyTransactionWriter
Session
-currentUser: User*
-loggedIn: bool
-creditAddedThisSession: double
+login(user: User*): bool
+logout(): bool
+isLoggedIn(): bool
+getCurrentUser(): User*
+isAdmin(): bool
+canSell(): bool
+canBuy(): bool
+addSessionCredit(amount: double): bool
+getSessionCredit(): double
AccountFile
-users: vector<User>
+load(path: string): bool
+findUser(username: string): User*
+userExists(username: string): bool
+addUser(user: User): void
+removeUser(username: string): bool
+getAllUsers(): vector<User>
GameFile
-available: vector<Game>
-pendingListings: vector<Game>
+load(path: string): bool
+findAvailable(name: string): Game*
+nameInUse(name: string): bool
+addPending(game: Game): void
+commitPending(): void
+removeBySeller(seller: string): void
+getAvailable(): vector<Game>
CollectionFile
-entries: vector<CollectionEntry>
+load(path: string): bool
+ownsGame(owner: string, game: string): bool
+addEntry(owner: string, game: string): void
Console
+readLine(line: string&): bool
+prompt(message: string): void
+showMessage(message: string): void
+showError(message: string): void
+showGameList(games: vector<Game>): void
+showAccountList(users: vector<User>): void
DailyTransactionWriter
-out: ofstream
+open(path: string): bool
+writeSessionEnd(user: User): void
+writeCreate(user: User): void
+writeDelete(user: User): void
+writeSell(game: Game): void
+writeBuy(game: Game, buyer: string): void
+writeRefund(buyer: string, seller: string, amount: double): void
+writeAddCredit(user: User): void
+close(): void
1
1
1
1
1
1
1
1
0..*
0..*
User
-username: string
-type: UserType
-credit: double
+getUsername(): string
+getType(): UserType
+getCredit(): double
+addCredit(amount: double): void
+deductCredit(amount: double): bool
+isAdmin(): bool
+canSell(): bool
+canBuy(): bool
«enumeration»
UserType
AA
FS
BS
SS
0..*
Game
-name: string
-sellerName: string
-price: double
+getName(): string
+getSeller(): string
+getPrice(): double
0..*
CollectionEntry
-ownerName: string
-gameName: string
+getOwner(): string
+getGameName(): string